

B.Tech. Degree IV Semester Supplementary Examination March 2024

ME 19-205-0403 MECHATRONICS
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

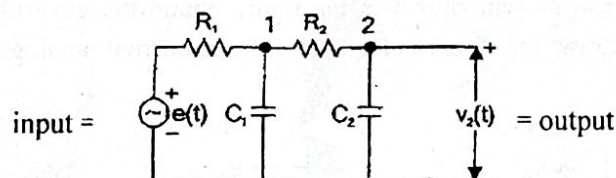
On successful completion of the course, the students will be able to:

- CO1: Understand the basics of sensors and transducers and different mechatronics systems.
 CO2: Model and analyse first and second order systems for their time and frequency domain responses.
 CO3: Analyse a system for its stability.
 CO4: Design mechatronics system for simple applications.
 CO5: Use software tools for modeling and simulation of mechatronics systems.
 Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
 PO – Programme Outcome

PART A

(Answer *ALL* questions)

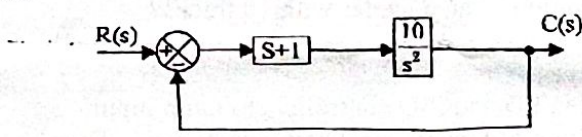
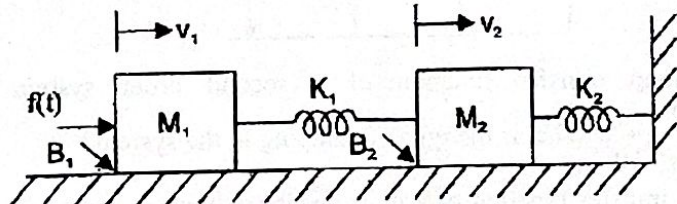
- | | | (8 × 3 = 24) | Marks | BL | CO | PO |
|--------|--|--------------|-------|----|----|-------|
| 1. (a) | Using a neat sketch describe an absolute optical encoder with two tracks. What is the resolution of an encoder with 10 tracks? | 3 | 3 | L2 | 1 | 1,2 |
| (b) | Draw the schematic representations for the different actuation methods for direction control valves. | 3 | 3 | L1 | 1 | 1,2 |
| (c) | Discuss the response of P, PD and PID controllers to ramp input. | 3 | 3 | L3 | 2 | 1,2 |
| (d) | Obtain the transfer function of the electrical network shown in figure. | 3 | 3 | L4 | 2 | 1,2,3 |



- | | | | | | | |
|-----|---|---|---|----|---|-------|
| (e) | The closed loop transfer function of a second order system $\frac{C(S)}{R(S)} = \frac{10}{S^2 + 6s + 10}$. What is the type of damping in the system? | 3 | 3 | L2 | 2 | 1,2,3 |
| (f) | The open loop transfer function of a unity feedback control system is given by $G(S) = \frac{K}{(S+2)(S+4)(S^2+6s+25)}$. By applying Routh – Hurwitz criteria, discuss the stability of the closed-loop system as a function of K. Determine the value of K which will cause sustained oscillations. | 3 | 3 | L3 | 3 | 1,2,3 |
| (g) | When an input X_1 is given, an output Y_1 is produced. After a delay of 5 seconds, another output Y_2 is also turned on. After another 15 seconds, both Y_1 and Y_2 are turned off. Draw a ladder diagram for this case. | 3 | 3 | L3 | 1 | 1,2,3 |
| (h) | Inputs A and B are connected to an AND gate. The output of the AND gate and input C are connected to a NOR gate. The NOR gate gives an output Y. Draw the circuit and the truth table. | 3 | 3 | L3 | 1 | 1,2,3 |

(P.T.O.)

PART B

		(4 × 12 = 48)	Marks	BL	CO	PO
II.	(a) Describe:		6	L1	1	1,2
	(i) LVDT					
	(ii) Strain Gauge					
	(iii) Vane Pump.					
	(b) Explain pulse width modulation. Represent a 30% duty cycle PWM signal.		3	L2	1	1,2
	(c) Describe the working of thermistors and RTDs.		3	L1	1	1,2
	OR					
III.	(a) Explain:		6	L1	1	1,2
	(i) 4/3 DCV					
	(ii) Hydraulic capacitance					
	(iii) Pneumatic proximity sensor.					
	(b) Discuss a few applications of mechanical linkages in automation.		3	L2	1	1,2
	(c) Explain the working of permanent magnet and variable reluctance type stepper motors.		3	L1	1	1,2
IV.	(a) For unity feedback system having open loop transfer function as $G(S) = \frac{K(S+2)}{S^2(S^2+7s+12)}$, determine:-		6	L3	3	1,2,3
	(i) Static error constants					
	(ii) Steady state error for parabolic input.					
	(b) Determine the time response of the system shown in the figure for unit step input.		6	L3	3	1,2,3
						
	OR					
V.	(a) For the mechanical system shown in the figure obtain the governing equations and draw the force-voltage and force-current analogous circuits.		6	L4	3	1,2,3
						
	(b) The unity feedback system is characterized by an open loop transfer function $G(S) = \frac{K}{S(S+10)}$. Determine the gain K, for which the system will have a damping ratio of 0.5. Determine peak overshoot and time at peak overshoot for a unit step input.		6	L4	3	1,2,3

(Continued)

		Marks	BL	CO	PO
VI.	Sketch the Bode plot for the following transfer function and determine the system gain K for the gain cross over frequency to be 5 rad/sec.	12	L4	3	1,2,3
	$G(S) = \frac{Ks^2}{(1+0.2s)(1+0.02s)}$				
	OR				
VII.	A unity feedback control system has an open loop transfer function, $G(S) = \frac{K}{S(S^2 + 4s + 13)}$. Sketch the root locus.	12	L4	3	1,2,3
VIII (a)	On receiving a signal from a sensor, a single acting pneumatic cylinder extends, remains extended for three seconds and then retracts. The cylinder is controlled by a two-solenoid 3/2 DCV. Draw the hydraulic/electric circuits and the required ladder diagram.	8	L4	1	1,2,3
(b)	Represent the following using logic gates:- (i) An output Y is generated when input A and one among inputs B and C are given. (ii) When inputs A and B are 1, output U is produced; A is 1 and B is 0, output V is produced; when both A and B are 0, output W is produced.	4	L4	1	1,2,3
	OR				
IX. (a)	Explain: (i) Accumulator Register (ii) Address Bus (iii) Flag Registers.	3	L1	1	1,2,3
(b)	Explain the working of a coin operated, PLC controlled car park barrier.	9	L2	1	1,2,3

Bloom's Taxonomy Levels

L1 - 22.5%, L2 - 17.5%, L3 - 20%, L4 - 40%.
